Lecture 8: Legal texts as data

Empirical Methods for Law and Economics | EDLE

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Today's questions

How do we turn a constitution into a number?

How do we know the number measures what we want?

- By the end of today you will be able to:
  - describe the three steps that turn text into data;
  - build a dictionary measure from legal texts in Stata;
  - validate it: read what it counts, look for what it misses, compare it with other measures;
  - revise your draft referee comments.

DISCUSS De jure independence, coded by hand

- Voigt and Wulf measure the *de jure* independence of prosecutors with up to 21 items, coded by reading legal texts for 76 countries (Lecture 4).

Questions:

1. What does hand-coding cost, and what can go wrong?
2. Could a computer read the texts instead? What would it do well, and what badly?

ANSWERS De jure independence, coded by hand

- Cost: expert time for every country and every revision of the law. Risks: coders who disagree, coders who know the answer they expect, rules that are hard to reproduce.
- A computer applies the same rule to every text, at no extra cost, and anyone can rerun it.
- But it reads words, not meaning: “independent” in a provision on courts is not “independent” in the name of a tribunal or in the independence of the state.
- Today: build such a measure, then check it as carefully as a hand-coded one.

Text as data in three steps

1. Represent raw text as numbers: for each document, the presence or count of words or phrases.
2. Map those numbers into a measure of what we care about.
3. Use the measure in a descriptive or causal analysis.

- Each step involves choices. The first: **what is a document?** A constitution, an article, a sentence?
- The simplest way to do step 2 is a **dictionary**: a list of words chosen by the researcher. Dictionary methods "do not involve statistical inference at all" and are "by far the most common method" in the social sciences.

Gentzkow, M., Kelly, B. and Taddy, M. (2019), *Journal of Economic Literature* 57(3): 535–574, pp. 536 and 541.

Our corpus

- The constitutions of 25 European countries, in English, from Constitute Project (CC BY-NC 3.0): 25 PDF files.
- Most are translations; versions differ: each text includes amendments up to a given year (Italy: 2012).
- A PDF is made to be read by people, not by a computer. Before counting anything we need:
  1. the text, without what is on the page for the reader only (**PDF to text**);
  2. one row per document, here one row per **sentence** (**text to data**).

Elkins, Z., Ginsburg, T. and Melton, J., *Constitute: The World's Constitutions to Read, Search, and Compare*, constituteproject.org. Details in `text_analysis/LICENCE_AND_SOURCE.txt`.

From PDF to text: what is on the page



- The program `pdftotext` gives every word with its position on the page and the size of its font. The rules use both.
- Also dropped: the cover page, the table of contents and the topic index.

Italy's Constitution of 1947 with amendments through 2012, p. 4, constituteproject.org.

From PDF to text: before and after

All the text on the page, line by line

```
constituteproject.org
PDF generated: 27 Apr 2022, 12:40
Art 8
All religious denominations are equally ...
Denominations other than Catholicism hav...
their own statutes, provided these do no...
Their relations with the State are regul...
respective representatives.
Right to culture
Art 9
Reference to art
Reference to science
The Republic promotes the development of...
research.
```

Long lines are cut at the right edge of the slide (...).

`Italy_2012.txt`: one sentence per line

```
All religious denominations are equally ...
Denominations other than Catholicism hav...
Their relations with the State are regul...
The Republic promotes the development of...
It safeguards natural landscape and the ...
```

- Header, headings and margin labels are gone; lines broken by the page layout are joined.
- A sentence ends with . ; or : followed by a capital letter or a number.
- Result: one text file per constitution.

From PDF to text: the script

- `text_analysis/08_pdf_to_txt.py` (Python; it uses the program `pdftotext`). The core rules, slightly simplified:

```
# keep a word only if its font is not small (margin labels)
# and it is not at the top or bottom of the page (headers, footers)
ws = [w for w in words(page) if height(w) > 9.5 and 35 < top(w) < 750]

# a line in a large font is a heading: it closes the previous sentence
kind = "H" if height >= 13.5 else "B"

# a sentence ends with . ; or : followed by a capital letter, a digit,
# a bracket or a quotation mark
re.split(r'(?<=[.;:])\s+(?=[A-ZO-9(\["])', text)
```

- Every threshold is a choice made by looking at the PDFs. Change it, and the data change.
- Output: `text_analysis/constitutions_txt/`, one file per constitution, one sentence per line, and `constitutions_list.csv`.

STATA Guided Stata 1: from text files to a data set

```
global text_dir "../../text_analysis/constitutions_txt"
import delimited using "$text_dir/constitutions_list.csv", ///
    varnames(1) encoding("utf-8") clear
save "$course_results/tables/08_constitutions_list.dta", replace
levelsof constitute_id, local(ids)
clear
tempfile corpus
save `corpus', emptyok
foreach id of local ids {
    import delimited text using "$text_dir/`id'.txt", ///
        delimiter(tab) varnames(nonames) bindquote(nobind) stripquote(no) ///
        stringcols(_all) encoding("utf-8") clear
    generate constitute_id = "`id'"
    generate sent_id = _n
    append using `corpus'
    save `corpus', replace
}
merge m:1 constitute_id using "$course_results/tables/08_constitutions_list.dta", nogenerate
sort constitute_id sent_id
order economy_code country constitute_id year_adopted year_text sent_id text
save "$course_data/constitutions_text.dta", replace
```

- Each line of a text file becomes a row; `foreach` repeats the import for the 25 files and `append` stacks them.
- Result: 18,853 rows, saved as `data/constitutions_text.dta`. The copy you received in `data/` is identical: if the loop fails, continue from Guided Stata 2.

STATA Guided Stata 2: the corpus

```
use "$course_data/constitutions_text.dta", clear
describe
generate words = wordcount(subinstr(text, char(34), "", .))
tabstat words, by(country) statistics(n sum)
```

Country	Sentences	Words
Sweden	2,190	59,285
Austria	1,795	49,362
	...	
Denmark	246	6,163
Latvia	219	4,962
All 25	18,853	466,841

- Lengths differ by a factor of 12. Any count must be compared with something.

PREDICT Who writes about the independence of judges?

Before any code: name two constitutions you expect to say a lot about the independence of judges, and two you expect to say little. Why?

- Think about when the constitution was written, and about what each country has had to protect its courts from.

A dictionary

- A sentence is about the **judiciary** if it contains: judge, judges, judicial, judiciary, court, courts, tribunal, tribunals.
- It mentions **independence** if it contains a word starting with: independen-, impartial-.
- Flag the sentences that do both.
- Measure for each country: the **share** of judiciary sentences that are flagged. A share, not a count, because constitutions differ in length.

STATA Guided Stata 3: applying the dictionary

```
generate text_lc = ustrlower(text)
generate byte judiciary = ustrregexm(text_lc, ///
    "\b(judge|judges|judicial|judiciary|court|courts|tribunal|tribunals)\b")
generate byte indep = ustrregexm(text_lc, "\b(independen|impartial)")
generate byte jud_indep = judiciary & indep
tabulate judiciary jud_indep
```

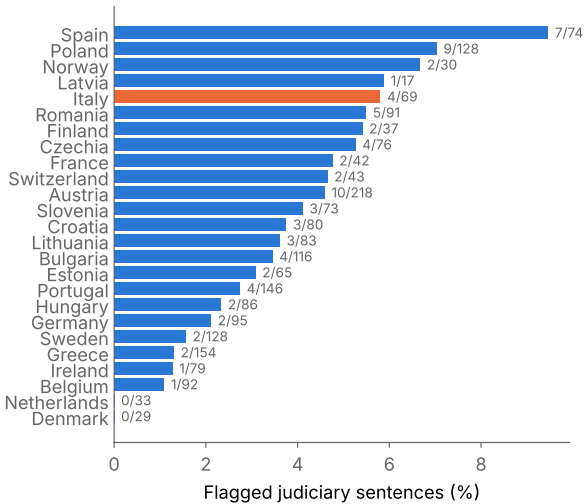
- `ustrregexm` returns 1 if the text matches a **regular expression**: `|` means "or", `\b` marks the edge of a word.
- Result: 2,084 sentences about the judiciary; 77 of them flagged.

STATA Guided Stata 4: one number per constitution

```
preserve
collapse (count) sentences=sent_id (sum) words judiciary jud_indep, ///
    by(economy_code country)
generate share = 100*jud_indep/judiciary
gsort -share
list country sentences words judiciary jud_indep share, noobs
graph hbar share, over(country, sort(1) descending)
save "$course_results/tables/08_constitutions_country.dta", replace
restore
```

- `collapse`: from 18,853 sentences to 25 constitutions (number of sentences; sums of words and of the two indicators).
- The country file is saved for later; `restore` brings back the sentences.

The measure



- Labels: flagged sentences / judiciary sentences.
- Spain and Poland at the top; the Netherlands and Denmark at zero.
- Italy: 4 of 69, 5.8%.

DISCUSS Reading the ranking

Questions:

1. Does Spain's constitution protect judicial independence more than any other?
2. The Netherlands and Denmark score zero. Do their judges lack independence?
3. Austria has 10 flagged sentences, more than any other country. What would you check first?

ANSWERS Reading the ranking

- The share counts words, not guarantees. A constitution can say “independent” many times and protect judges little, or say it once and protect them well.

- A zero means that these words do not appear in sentences about courts. It says nothing yet about the practice, and perhaps not even about the text: the provision may use other words.

- Check what was counted: read the 10 Austrian sentences. Before trusting a measure, read what it measures.

Validation

- Gentzkow, Kelly and Taddy (Section 3.5.2): a manual audit, "cross-checking some subset of the fitted values against the coding a human would produce"; "often as few as twenty or thirty documents is enough". Auditing "is especially important for dictionary methods".
- Two questions:
  - **Precision**: of the sentences we flagged, how many are really about the independence of courts?
  - **Recall**: of the sentences that are about it, how many did we flag?
- Then compare the measure with other measures of the same concept.

Gentzkow, Kelly and Taddy (2019), p. 555.

STATA Guided Stata 5: read what you counted

```
set seed 20261030
generate double u = runiform()
gsort -jud_indep u
list country text in 1/20, noobs
```

- `runiform()` gives each sentence a random number; `gsort` puts the flagged sentences first, in random order. The first 20 rows are a random sample of flagged sentences.
- Code each one: is it about the independence or impartiality of courts or judges? Yes or no.

DISCUSS Six flagged sentences

1. Italy: "The Judiciary is a branch that is autonomous and independent of all other powers."
2. Austria: "Per 1. July 2008 the former independent Federal Asylum Tribunal becomes the Asylum Court."
3. Germany: "The Federal Court of Audit, whose members shall enjoy judicial independence, shall audit the account [...]"
4. France: "Statutes shall determine the rules concerning: civic rights [...]; freedom, diversity and the independence of the media; [...]"
5. Hungary: "Everyone shall have the right to have any charge against him or her [...] adjudicated [...] in a fair and public trial by an independent and impartial court established by an Act."
6. Croatia: "The Public Prosecution Service is an autonomous and independent judicial body [...]"

Correct or not? For each error, say which word misled the dictionary.

ANSWERS Six flagged sentences

- Correct: (1) Italy and (5) Hungary, a right to an independent court.
- Errors: (2) “independent” is part of the name of a tribunal; (3) a court of audit, not a court of justice; (4) “independence” refers to the media, and the judiciary word is “courts” in “new categories of courts”, later in the same long sentence; (6) prosecutors, not judges.
- I read all 77 flagged sentences: 15 are wrong, so **precision is 81%**. Nine of the 15 are Austrian transitional provisions: seven on the “independent Federal Asylum Tribunal”, two on the “Independent Administrative Authorities”. Austria has 1 correct sentence, not 10.
- Errors are not random: they concentrate in one country and change its rank. An average precision hides that.

What the dictionary misses

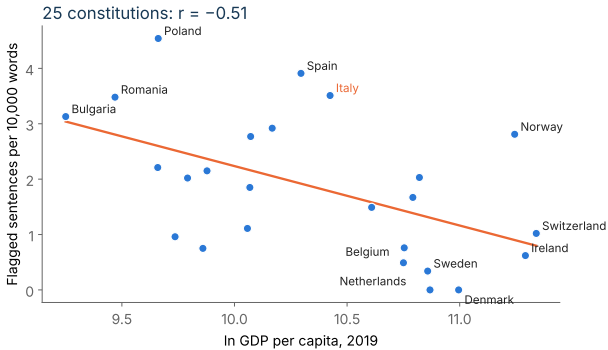
- Denmark, section 64: “In the performance of their duties the judges shall be directed solely by the law.”
- Italy, Article 101: “Judges are subject only to the law.”
- France, Article 64: “Judges shall be irremovable from office.”
- Same concept, other words: low **recall**. A second dictionary adds *irremovab-*, “subject only to”, “bound only by”, “solely by the law”.
- Version 2: 86 flagged sentences instead of 77. Denmark $0 \to 1$, Italy $4 \to 5$, France $2 \to 3$, Bulgaria $4 \to 7$.
- Every new word adds recall and may cost precision: each version must be audited again.

STATA Guided Stata 6: compare with other measures

```
use "$course_results/tables/08_constitutions_country.dta", clear
merge 1:1 economy_code using "$course_data/justice_europe_2019.dta", ///
    keep(match) nogenerate
generate per10k = 10000*jud_indep/words
generate ln_gdppc = ln(gdp_pc_usd)
pwcorr share judicial_quality, sig obs
pwcorr per10k ln_gdppc, sig obs
```

- `keep(match)` keeps the 25 countries found in both files; `nogenerate` skips the `_merge` variable.
- Share and the Doing Business index of the quality of judicial processes: $r = 0.20$ ($p = 0.33$, 25 countries).
- Flagged sentences per 10,000 words and ln GDP per capita: $r = -0.51$ ($p = 0.009$).

Text and income



- Poorer and newer democracies write more about independent judges; the Netherlands, Denmark, Sweden and Belgium write least.
- With the share instead of the per-word measure the correlation is weaker: $r = -0.22$ ($p = 0.30$). And neither measure removes the Austrian errors. A result that depends on the choice of measure must be reported with both.

DISCUSS Richer countries write less about independent judges

Questions:

1. How would you explain the negative correlation with income?
2. Does it show that the text measure is invalid?
3. What does it remind you of in Voigt and Wulf?

ANSWERS Richer countries write less about independent judges

- Newer constitutions, written after 1990 in countries that had experienced political control of courts, spell out guarantees; older ones in stable democracies (Netherlands, Denmark, Belgium) rely on statutes and practice.

- Not invalid, but it measures *what is written*, not *what happens*. Both are legitimate objects; they must not be confused.

- Voigt and Wulf: *de jure* and *de facto* independence of prosecutors are weakly negatively correlated ($r = -0.21$, Van Aaken et al. 2010, cited by Voigt and Wulf). Law on paper and law in practice can diverge.

- A referee would ask: which concept does the paper need, and is the text measure validated against it?

A referee's checklist for text-based measures

- **Documents**: which texts, which version, which language? Translations are not the legal text (the Swiss English version "has no legal force").
- **Unit**: sentence, article, document? Long texts produce more matches.
- **Dictionary**: who chose the words, before or after seeing the results?
- **Audit**: is there a reading of a random sample, with the share of errors? By country?
- **Recall**: what provisions use other words?
- **Concept**: text or practice? Is the measure compared with an independent measure of the same concept?
- **Processing**: extraction from PDFs leaves errors (one German sentence begins "5. state security." from the line above).

Your draft comments

- Check each comment against two lists.

The four steps (Lecture 6)

1. Where: table, column, page.
2. What: a fact that can be checked.
3. Why it matters: which conclusion.
4. What would resolve it.

Berk et al. (Lecture 5)

- Essential or suggestion? Labelled?
- Is the argument scientific, not a hunch?
- Is the request worth its cost?
- Is it about the paper the authors wrote?

DISCUSS Revising a comment

Read one of your comments aloud. The class says:

1. which of the four steps is missing;
2. whether it is essential or a suggestion;
3. one way to make it more specific.

- Then rewrite your weakest comment.

ANSWERS Revising a comment: frequent problems

- No location: “the identification is weak” without the table or the assumption at stake.
- No consequence: the issue is real, but which conclusion would change?
- No way out: ask for a check the authors can run, or for a narrower claim.
- A suggestion presented as essential, or a different paper: label it, or drop it.
- Tone: write about the paper, not about the authors.

Laboratory

- Worksheet: `do/exercises/08_text_as_data.do`.
- Data: `constitutions_text.dta`.
- Tasks: audit 20 flagged sentences; find what the dictionary misses in two constitutions; build and audit the second dictionary; prosecutors instead of judges; two ways to normalise; one written comment.
- Then: finish the revision of your comments.

Exit check

1. What are the three steps that turn text into data?
2. Define precision and recall in one sentence each.
3. Give one reason why a constitution can score zero on our measure and still protect the independence of judges.

Next lectures

- Lectures 9–10: your presentations, each followed by questions from the class (format in the syllabus).
- Present the question, one central result, a strength and your most important comment.
- Ask at least one question in each presentation other than your own: your best three questions count.
- Final report: by the common deadline announced on Teams.